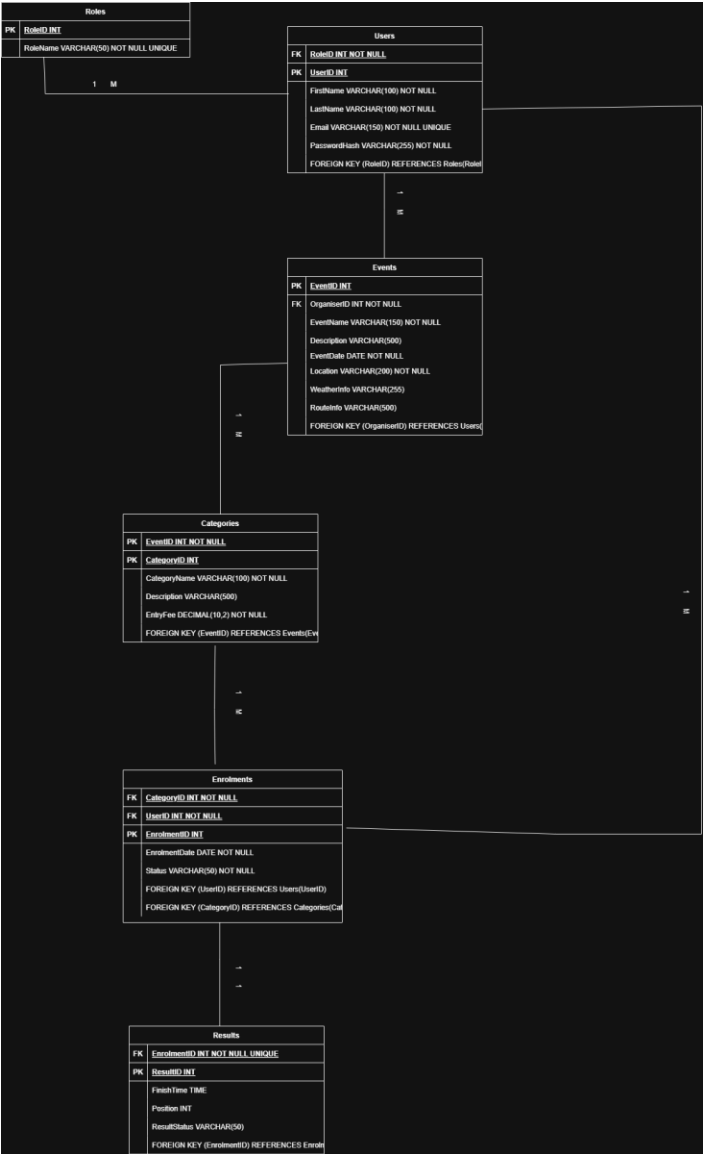


SECTION B: ENTITY RELATIONSHIP DIAGRAM (ERD)

System Overview

The RaceDay system is a full-stack web-based event management platform for the South African road running, walking, and cycling community. The database consists of six interconnected entities that support role-based access control, event management, participant enrolments, and race result tracking.

[SCREENSHOT 2: ERD Diagram]



ERD Description

The ERD contains six entities with the following design rationale:

Roles: Stores user role definitions (Organiser, Participant, Admin). RoleName is unique to prevent duplicate roles.

Users: Stores registered users. RoleID is a foreign key linking to Roles. Email is unique to prevent duplicate accounts. PasswordHash stores encrypted passwords for security.

Events: Stores race events created by organisers. OrganizerID is a foreign key to Users. Includes EventName, Description, EventDate, Location, MaxEntrants, and PostCode.

Categories: Stores event categories (e.g., Full Marathon, Half Marathon). EventID is a foreign key to Events. EntryFee is DECIMAL(10,2) for accurate currency representation.

Enrolments: Junction table linking Users to Categories and Events. Tracks EnrolmentDate and Status (Confirmed, Pending, Cancelled). Contains three foreign keys.

Results: Stores race outcomes. EnrolmentID is both a foreign key and UNIQUE, enforcing a strict one-to-one relationship — one enrolment has exactly one result.

SECTION B: DATA DICTIONARY

The following table details every column in the database schema, including data types, constraints, and descriptions.

Table	Column	Data Type	Constraints	Description
Roles	RoleID	INT	PK, IDENTITY(1,1)	Unique identifier for each role
Roles	RoleName	VARCHAR(50)	NOT NULL, UNIQUE	Role name: Organiser, Participant, Admin
Users	UserID	INT	PK, IDENTITY(1,1)	Unique identifier for each user
Users	RoleID	INT	FK, NOT NULL	Links user to their role in Roles table
Users	FirstName	VARCHAR(100)	NOT NULL	User first name
Users	LastName	VARCHAR(100)	NOT NULL	User last name
Users	Email	VARCHAR(100)	NOT NULL, UNIQUE	User email address (login credential)
Users	PasswordHash	VARCHAR(255)	NOT NULL	Encrypted password for authentication
Events	EventID	INT	PK, IDENTITY(1,1)	Unique identifier for each event
Events	OrganizerID	INT	FK, NOT NULL	Links event to the organiser who created it
Events	EventName	VARCHAR(150)	NOT NULL	Name of the race event
Events	Description	VARCHAR(500)	NULL	Detailed description of the event
Events	EventDate	DATE	NOT NULL	Date when the event takes place
Events	Location	VARCHAR(100)	NOT NULL	Physical location of the event
Events	MaxEntrants	VARCHAR(50)	NULL	Maximum number of allowed participants
Events	PostCode	VARCHAR(20)	NULL	Postal code for the event location
Categories	CategoryID	INT	PK, IDENTITY(1,1)	Unique identifier for each category
Categories	EventID	INT	FK, NOT NULL	Links category to its parent event
Categories	CategoryName	VARCHAR(100)	NOT NULL	Name of the category (e.g., Full Marathon)
Categories	Description	VARCHAR(500)	NULL	Description of the category
Categories	EntryFee	DECIMAL(10,2)	NOT NULL	Cost to enter this category (ZAR)
Enrolments	EnrolmentID	INT	PK, IDENTITY(1,1)	Unique identifier for each enrolment
Enrolments	CategoryID	INT	FK, NOT NULL	Links to the

				category being entered
Enrolments	UserID	INT	FK, NOT NULL	Links to the participant who enrolled
Enrolments	EventID	INT	FK, NOT NULL	Links to the event being entered
Enrolments	EnrolmentDate	DATE	NOT NULL	Date when the enrolment was made
Enrolments	Status	VARCHAR(50)	NOT NULL	Enrolment status: Confirmed, Pending, Cancelled
Results	ResultID	INT	PK, IDENTITY(1,1)	Unique identifier for each result
Results	EnrolmentID	INT	FK, NOT NULL, UNIQUE	Links to enrolment (one result per enrolment)
Results	FinishTime	TIME	NULL	Participant finish time (HH:MM:SS)
Results	Position	INT	NULL	Finishing position in the race
Results	ResultStatus	VARCHAR(20)	NULL	Result status: Completed, DNF, Disqualified

SECTION B: API ENDPOINT PLAN

The following table lists all 22 planned API endpoints for the RaceDay system. This plan was completed before any application code was written and covers Authentication, User Profiles, Events, Categories, Enrolments, and Results.

Authentication Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user with role assignment	None (public)	{ firstName, lastName, email, password, roleName }	201 Created — User object with token. 400 — Invalid data. 409 — Email exists.
POST	/api/auth/login	Authenticates user and returns JWT token	None (public)	{ email, password }	200 OK — JWT token + user details. 401 — Invalid credentials.

User Profile Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/profile	Retrieves logged-in user profile	Any (logged in)	None	200 OK — User profile. 401 — Not logged in.
PUT	/api/users/profile	Updates user profile	Any (logged in)	{ firstName, lastName, email, passwordHash }	200 OK — Updated profile. 400 — Invalid data.
GET	/api/users/{id}/enrolments	Gets all enrolments for a user	Any (own only)	None	200 OK — List of enrolments. 403 — Accessing other data.
GET	/api/users/{id}/results	Gets all race results for a user	Any (own only)	None	200 OK — List of results. 403 — Accessing other data.

Event Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists all upcoming events	None (public)	None	200 OK — Array of events. 404 — No events.
GET	/api/events/{id}	Gets details for a specific event	None (public)	None	200 OK — Event + categories. 404 — Not found.
POST	/api/events	Creates a new	Organiser	{ eventName,	201 Created —

		event		description, eventDate, location, maxEntrants, postCode }	New event. 400 — Missing fields. 403 — Not organiser.
PUT	/api/events/{id}	Updates an existing event	Organiser (owner)	{ eventName, description, eventDate, location, maxEntrants, postCode }	200 OK — Updated event. 403 — Not owner. 404 — Not found.
DELETE	/api/events/{id}	Deletes an event	Organiser (owner)	None	204 No Content. 403 — Not owner. 404 — Not found.
GET	/api/events/organiser/{id}	Gets all events by an organiser	Organiser	None	200 OK — Array of events. 403 — Not the organiser.

Category Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{id}/categories	Gets all categories for an event	None (public)	None	200 OK — Array of categories. 404 — Event not found.
POST	/api/events/{id}/categories	Adds a category to an event	Organiser (owner)	{ categoryName, description, entryFee }	201 Created — New category. 400 — Invalid data. 403 — Not owner.
PUT	/api/categories/{id}	Updates a category	Organiser (owner)	{ categoryName, description, entryFee }	200 OK — Updated category. 403 — Not owner. 404 — Not found.
DELETE	/api/categories/{id}	Deletes a category	Organiser (owner)	None	204 No Content. 403 — Not owner.

Enrolment Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments	Registers participant into a category	Participant	{ categoryId, eventId, enrolmentDate }	201 Created — Enrolment. 409 — Already enrolled. 403 — Not participant.
GET	/api/enrolments	Gets all enrolments (filtered by role)	Any (logged in)	None	200 OK — Array of enrolments.
GET	/api/enrolments/{id}	Gets a specific enrolment	Any (owner or organiser)	None	200 OK — Enrolment details. 403 — No access. 404 — Not found.

PUT	/api/enrolments/{id}/status	Updates enrolment status	Organiser	{ status }	200 OK — Updated. 403 — Not organiser.
DELETE	/api/enrolments/{id}	Cancels an enrolment	Participant (own) or Organiser	None	204 No Content. 403 — No access.

Results Endpoints

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Records a race result	Organiser	{ enrolmentId, finishTime, position, resultStatus }	201 Created — Result. 400 — Invalid data. 403 — Not organiser.
GET	/api/results	Gets results (filtered by role)	Any (logged in)	None	200 OK — Array of results.
GET	/api/results/{id}	Gets a specific result	Any (owner or organiser)	None	200 OK — Result details. 403 — No access. 404 — Not found.
GET	/api/events/{id}/results	Gets leaderboard for an event	None (public)	None	200 OK — Results sorted by position. 404 — Event not found.
PUT	/api/results/{id}	Updates a result	Organiser	{ finishTime, position, resultStatus }	200 OK — Updated. 403 — Not organiser. 404 — Not found.

Total Endpoints Planned: 22

Coverage: Authentication (2), Users (4), Events (6), Categories (4), Enrolments (5), Results (5)

SECTION B: ENTITY RELATIONSHIPS

The following diagram and descriptions explain the cardinality between all entities in the RaceDay database.

Relationship Descriptions

Roles → Users: One-to-Many (1:∞)

One role can be assigned to many users. For example, the 'Participant' role can have hundreds of users assigned to it. The RoleID in Users is a foreign key referencing Roles(RoleID).

Users → Events: One-to-Many (1:∞)

One organiser (user with Organiser role) can create many events. The OrganizerID in Events is a foreign key referencing Users(UserID). An event can only have one organiser.

Events → Categories: One-to-Many (1:∞)

One event can have multiple categories. For example, the Soweto Marathon has Full Marathon, Half Marathon, and 10km Fun Run categories. The EventID in Categories is a foreign key referencing Events(EventID).

Users → Enrolments: One-to-Many (1:∞)

One participant can enrol in multiple events over time. The UserID in Enrolments is a foreign key referencing Users(UserID).

Categories → Enrolments: One-to-Many (1:∞)

One category can have many enrolments. The CategoryID in Enrolments is a foreign key referencing Categories(CategoryID).

Events → Enrolments: One-to-Many (1:∞)

One event can have many enrolments. The EventID in Enrolments is a foreign key referencing Events(EventID).

Enrolments → Results: One-to-One (1:1)

One enrolment has exactly one result. The EnrolmentID in Results is both a foreign key and has a UNIQUE constraint, enforcing this strict one-to-one relationship.

[SCREENSHOT 3: Database Diagram from SSMS]

The screenshot displays the SQL Server Enterprise Manager (SSMS) interface. The left pane shows the Object Explorer with the database structure of 'ApexCircuitDB'. The central pane shows a SQL query window with the following query:

```
1 SELECT * FROM Roles;
2 SELECT * FROM Users;
3 SELECT * FROM Categories;
4 SELECT * FROM Results;
```

The right pane shows the Solution Explorer with a list of miscellaneous files, including 'SQLQuery10.sql' through 'SQLQuery9.sql'.

Below the query window, the 'Results' tab is active, displaying a table with 4 columns: 'ResultID', 'EnrolmentID', 'FinishTime', and 'Position'. The table contains 4 rows of data:

ResultID	EnrolmentID	FinishTime	Position
1	1	02:45:30.0000000	15
2	2	01:30:15.0000000	42
3	3	03:15:00.0000000	8
4	4	05:30:45.0000000	120

The status bar at the bottom indicates 'Query executed succ...' and 'LabVM2049939:SQLEXPRESS (17...)'.